Integrated Project: Understanding and trusting data (Part 2). [MCQ] (Version : 0)

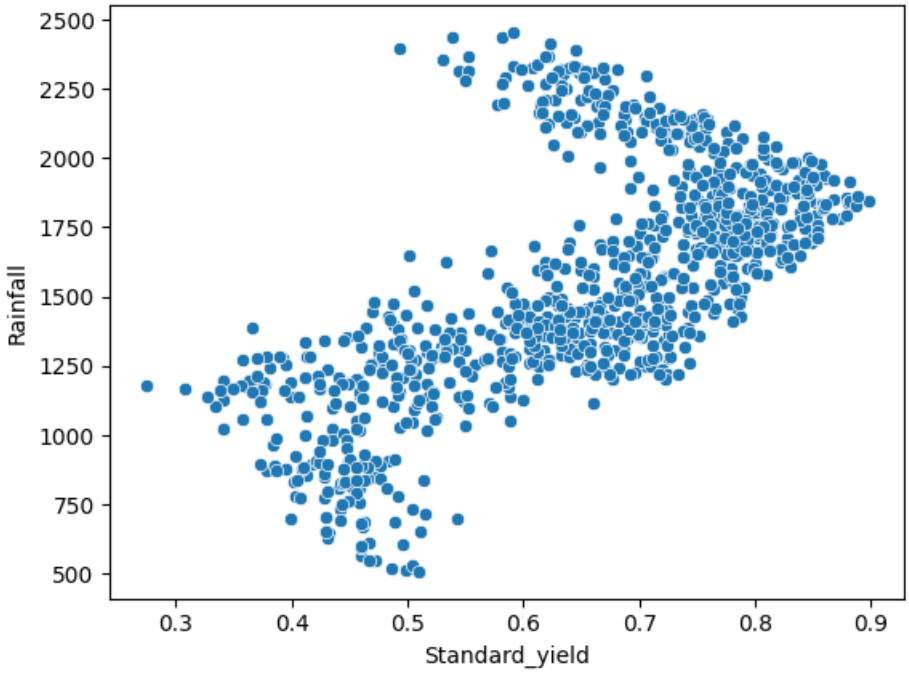
TEST

Correct Answer

Answered in 7.58 Minutes

Question 1/10

Based on the plot, which of the following best describes the pattern observed?



- [ ] The data indicates a weak linear relationship, with 'Standard_yield' slightly increasing as 'Rainfall' increases.
- [x] The data indicates a non-linear relationship, with 'Standard_yield' showing variability at different levels of 'Rainfall'.
- [ ] The data indicates no discernible relationship, with 'Standard_yield' being independent of 'Rainfall'.

- [ ] The data indicates a strong linear relationship, with 'Standard_yield' increasing consistently as 'Rainfall' increases.

Explanation:

The data indicates a strong linear relationship, with 'Standard_yield' increasing consistently as 'Rainfall' increases is incorrect because the relationship is not consistently increasing and does not fit a straight line.

The data indicates a weak linear relationship, with 'Standard_yield' slightly increasing as 'Rainfall' increases is incorrect because while there is a slight increase, calling the relationship weakly linear does not capture the complexity of the variability observed.

The data indicates no discernible relationship, with 'Standard_yield' being independent of 'Rainfall' is incorrect because there is a discernible pattern indicating some degree of dependency between 'Rainfall' and 'Standard_yield', even if it's not linear.

Question 2/10

Which weather station is linked to the most number of fields?

- [ ] 4
- [ ] 3
- [x] 0
- [ ] 1
- [ ] 2

Explanation:

1 is incorrect because Station 0 is linked to 1375 fields while Station 1 is only linked to 940 fields.

2 is incorrect because Station 0 is linked to 1375 fields while Station 2 is only linked to 1021 fields.

3 is incorrect because Station 0 is linked to 1375 fields while Station 1 is only linked to 1196 fields.

4 is incorrect because Station 0 is linked to 1375 fields while Station 1 is only linked to 1122 fields.

Question 3/10

An environmental monitoring station has updated its rainfall data transmission format. The current regex pattern for capturing rainfall data is not compatible with this new format. Given the existing pattern:

'Rainfall': r'(\d+(\.\d+)?)\s?mm',

The new message format is as follows:

"Rainfall Update: In the last 24 hours, there has been rainfall amounting to 25 millimetres as of 2024-01-31 10:00."

Which of the following modifications to the regex pattern will accurately capture the rainfall data in both the new and existing message formats?

- [ ] r'In the last 24 hours, (\d+(\.\d+)?)'
- [x] r'(\d+(\.\d+)?)\s?(mm|millimetres)'
- [ ] r'rainfall\samounting\s*to\s(\d+(\.\d+)?)'
- [ ] r'(\d+(\.\d+)?)\s?
- [ ] r'(\d+(\.\d+)?)\s*millimetres'

Explanation:

"r'(\d+(.\d+)?)\s*millimetres'" is incorrect because it specifically requires the word "millimetres" right after the number, which does not match the original message formats.

"r'rainfall\samounting\s*to\s(\d+(.\d+)?)'" is incorrect as it is tailored to a specific phrase structure that is not present in all message formats.

"r'In the last 24 hours, (\d+(.\d+)?)'" is incorrect because it relies on a specific preceding phrase, which limits its ability to match different message formats.

While this regex pattern will capture the rainfall data, it will also capture all other numerical characters, making the extraction of the rainfall data from the new message more complex.

Question 4/10

Which of the following crops have a strong positive linear relationship between Standard_Yield and pH?

- [x] Cassava
- [ ] Tea
- [ ] Potato
- [ ] Banana

Explanation:

Tea is incorrect because the correlation coefficient is only 0.05.

Potato is incorrect because the correlation coefficient is only 0.12.

Banana is incorrect because the correlation coefficient is - 0.37.

Question 5/10

What is the output of the following code?

```
def function(input):
    filtered_df = MD_agric_df[MD_agric_df['Crop_type'] == input]
    fig = px.scatter(filtered_df, x='Elevation', y='Annual_yield', color='Crop_type')
```

```
fig.show()

interact(function, input=MD_agric_df['Crop_type'].unique())
```

- [ ] An interactive scatter plot showing 'Elevation' vs 'Annual_yield' with different crop types displayed by colour.
- [ ] An error message due to incorrect use of the interact function.
- [x] An interactive scatter plot showing 'Elevation' vs 'Annual_yield' for the selected crop type, with a dropdown to filter by different 'Crop_type'.
- [ ] A series of static scatter plots for each 'Crop_type', showing 'Elevation' vs 'Annual_yield', displayed all at once.

Explanation:

An interactive scatter plot showing 'Elevation' vs 'Annual_yield' with different crop types displayed by colour is incorrect because the plot is not for all crop types simultaneously; it only shows data for the selected crop type from the dropdown.

A series of static scatter plots for each 'Crop_type', showing 'Elevation' vs 'Annual_yield', displayed all at once is incorrect as the code does not create static plots or display them all at once. Instead, it generates an interactive plot based on the selected input from the dropdown.

An error message due to incorrect use of the interact function is incorrect as the code is syntactically correct and uses the interact function appropriately. There should be no error message assuming MD_agric_df is a properly formatted DataFrame and Plotly and ipywidgets are correctly imported.

Question 6/10

Which variable is least correlated with Standard_yield?

- [ ] Pollution_level
- [ ] Standard_yield
- [ ] Annual_yield
- [x] Ave_temps

Explanation:

Pollution_level is incorrect because it is the strongest negatively correlated feature with Standard_yield.

Standard_yield is incorrect because it is always perfectly correlated with itself.

Annual_yield is incorrect because it has the strongest correlation with Standard_yield.

Question 7/10

Which of the following variables has the strongest linear relationship?

- [x] Rainfall and Soil_fertility
- [ ] Ave_temps and Soil_fertility
- [ ] Rainfall and Standard_yield
- [ ] Pollution_level and pH

Explanation:

Pollution_level and pH is incorrect because the correlation coefficient is only -0.141.

Rainfall and Standard_yield is incorrect because

the correlation coefficient is only 0.039.

Ave_temps and Soil_fertility is incorrect because the correlation coefficient is only 0.187.

Question 8/10

Create a KDE plot of the rainfall distribution and split the data by soil types.

What can we infer from this graph?

- [ ] Rocky, Sandy and Loamy soil types occur in places with rainfall > 1500 mm.
- [x] Rocky, Sandy and Loamy soil types occur in places with rainfall < 1500 mm.
- [ ] The distribution of rainfall in Peaty soils spans a range of more than 1500 mm.
- [ ] The maximum height of the Silty soil type distribution curve indicates that it is the most common soil type.

Explanation:

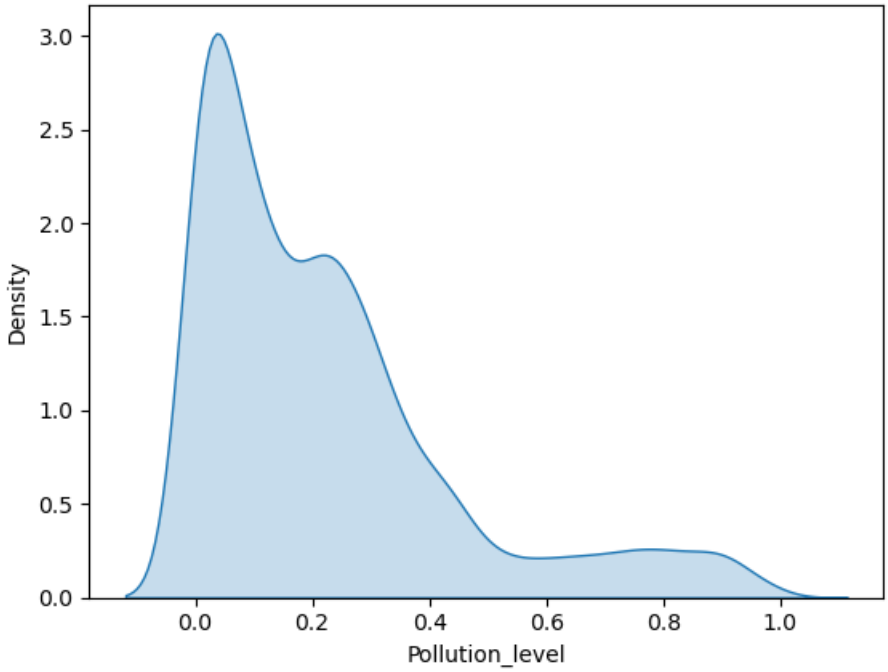
Rocky, Sandy and Loamy soil types occur in places with rainfall > 1500 mm is incorrect because those soil types occur in places with high rainfall.

The distribution of rainfall in Peaty soils only spans a range of approximately 500 mm.

The maximum height of the Silty soil type distribution curve indicates that it is the most common soil type is incorrect because the height of KDE plots does not infer the count.

Question 9/10

What can be inferred from the following plot?



- [ ] The maximum pollution level is 3.
- [x] Most of the fields in Maji Ndogo have a pollution rating less than 0.5.
- [ ] The plot indicates an error in the data.
- [ ] The mean is smaller than the median.
- [ ] Most of the fields in Maji Ndogo have a pollution rating less than 3.

Explanation:

Most of the fields in Maji Ndogo have a pollution rating of less than 3 is incorrect because the Density value describes how many data points have similar values.

The mean is smaller than the median is incorrect because the median is 0.16 while the mean is 0.22.

The maximum pollution level is 3 is incorrect because the height of the KDE plot does not describe the pollution levels.

The plot indicates an error in the data is incorrect

because while a skewed distribution needs extra care, the values are equally as valid.

Question 10/10

Why is some of the data in the .describe() method in Pandas not meaningful?

- [x] For columns like Field_ID or Crop_type, which contain unique identifiers or categorical labels, the computed descriptive statistics such as mean and median do not hold practical significance.
- [ ] The method is limited in processing large datasets, potentially leading to less meaningful outputs for columns with extensive data.
- [ ] There's a programming constraint in Pandas that prevents accurate calculation of descriptive statistics for certain column types.
- [ ] The method is not optimised for columns with mixed data types, leading to less meaningful or skewed statistical outputs.

Explanation:

The limitation in processing large datasets is incorrect because Pandas' .describe() method is capable of handling large datasets effectively, and the size of the dataset does not inherently affect the meaningfulness of the descriptive statistics.

The programming constraint in Pandas is incorrect because Pandas does not have such constraints. It can calculate descriptive statistics accurately for the data types it supports.

The method not being optimised for columns with mixed data types is incorrect because Pandas' .describe() method can handle columns with mixed data types, though the user must be cautious about the interpretation of the statistics in such cases.